
AGS: AGENTIC FRAMEWORKS

NewsGenie – An AI-Powered Information and News Assistant

Final Report

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Section 1: Executive Summary

Introduction

NewsGenie is an AI-powered News Intelligence Assistant designed to help users efficiently discover, understand, analyse and interact news through natural language conversations.

In today's information-rich environment, users are inundated with news from multiple sources, making it increasingly difficult to identify relevant developments, separate signal from noise, and quickly understand the implications of emerging events. Traditional news platforms require users to manually search, compare articles, and synthesize information from multiple sources, resulting in a fragmented and time-consuming experience.

To address this challenge, NewsGenie leverages an Agentic AI architecture that combines intelligent query understanding, dynamic workflow orchestration, real-time news retrieval and conversational response generation. The solution automatically distinguishes between news-related and general informational queries, retrieves relevant information from trusted external sources, maintains conversational context; and delivers concise, user-friendly insights.

Built using LangGraph, OpenAI, NewsAPI, Tavily and Gradio; NewsGenie demonstrates how multiple specialized AI agents can collaborate to create an intelligent, context-aware information assistant capable of transforming raw news into actionable insights.

Key outcomes delivered by the solution include:

- Faster access to relevant news and information.
- Reduced information overload through intelligent summarization.
- Improved user experience through conversational interactions.
- Context-aware follow-up handling across multi-turn conversations.
- Resilient operation through robust error handling and fallback mechanisms.

Business Context

The volume of digital information available to users has grown exponentially over the past decade. News is generated continuously across multiple platforms, publishers and social channels, creating significant challenges for users attempting to stay informed and make timely decisions.

While access to information has improved, the ability to efficiently consume and interpret that information has not kept pace.

Common challenges include:

- Information overload caused by continuous news generation.
- Difficulty identifying the most relevant developments within a topic.
- Fragmented information spread across multiple sources.
- Time-consuming manual research and validation efforts.
- Limited ability to ask follow-up questions or explore topics conversationally.
- Lack of contextual summaries that explain why developments matter.

These challenges create a gap between information availability and information usability.

Organizations and individuals increasingly require intelligent systems capable of not only retrieving information but also understanding intent, synthesizing content, maintaining context and presenting insights in a concise and actionable manner.

This creates an ideal use case for Agentic AI, where multiple specialized agents collaborate to perform classification, retrieval, enrichment, reasoning and response generation tasks dynamically based on user needs.

Problem Statement

The objective of this project is to design and implement an intelligent conversational assistant capable of helping users discover, understand and explore news content through natural language interactions.

The solution needed to address several key challenges:

Information Discovery Challenges

- Users struggle to locate relevant news across multiple sources.
- News searches often return excessive information with limited prioritization.
- Users spend significant time reviewing multiple articles to understand a topic.

Context and Interaction Challenges

- Traditional news platforms do not support conversational exploration.
- Follow-up questions often require users to repeat context.
- Users cannot easily transition between news retrieval and general knowledge questions.

Intelligence and Interpretation Challenges

- Raw articles frequently lack summarized context.
- Users need assistance understanding why a development is important.
- Related articles must be consolidated into a coherent narrative.

Operational Challenges

- External APIs and services may fail or become unavailable.
- User requests can be ambiguous or underspecified.
- The system must provide graceful recovery and meaningful guidance during failures.

The desired outcome is an intelligent assistant capable of functioning as a personalized news analyst—one that could understand user intent, retrieve relevant information, maintain conversational context and deliver concise, trustworthy and actionable responses.

Solution Overview

- To address above challenges, NewsGenie has been developed as an Agentic AI solution built on LangGraph, leveraging a collection of specialized agents that collaborate to process and respond to user requests.
- Rather than treating every query identically, the solution first analyses the user's request to determine:
 - Whether the user is seeking current news or general information.
 - Whether the query represents a new topic or a continuation of an existing conversation.

- Based on this understanding, the system dynamically routes requests through the appropriate workflow: -
 - For news-related queries, the solution retrieves relevant information from external news and web-search providers, enriches the results, consolidates overlapping information and generates an executive-style summary tailored to the user's request.
 - For general informational queries, the solution leverages the language model powered with web-search to provide direct responses while maintaining conversational continuity.
- The architecture incorporates:
 - Intelligent query classification and routing.
 - Context-aware conversational memory.
 - Persistent conversation management.
 - Multi-source information retrieval.
 - News enrichment and synthesis.
 - Structured response generation.
 - Service-aware error analysis and fallback handling.
 - Interactive Gradio-based user interface.
- By combining deterministic workflow orchestration with LLM-powered reasoning, NewsGenie delivers a scalable, extensible and user-centric framework for intelligent news discovery and conversational information retrieval.

Section 2: System Architecture

Introduction

- NewsGenie is an Agentic AI solution built on LangGraph, where specialized agents collaborate to classify queries, retrieve information, generate responses and handle failures.
- The solution follows a stateful, workflow-driven architecture that dynamically routes each request based on query type and user intent, ensuring the most appropriate processing path.
- Key architectural benefits include:
 - Agent specialization and clear separation of responsibilities
 - Dynamic workflow routing
 - Stateful multi-turn conversations
 - Robust error handling and recovery
 - Improved scalability and maintainability
- The architecture combines LLM-powered reasoning with external information sources (NewsAPI and Tavily) to deliver context-aware responses enriched with real-time information.

Architecture Diagram

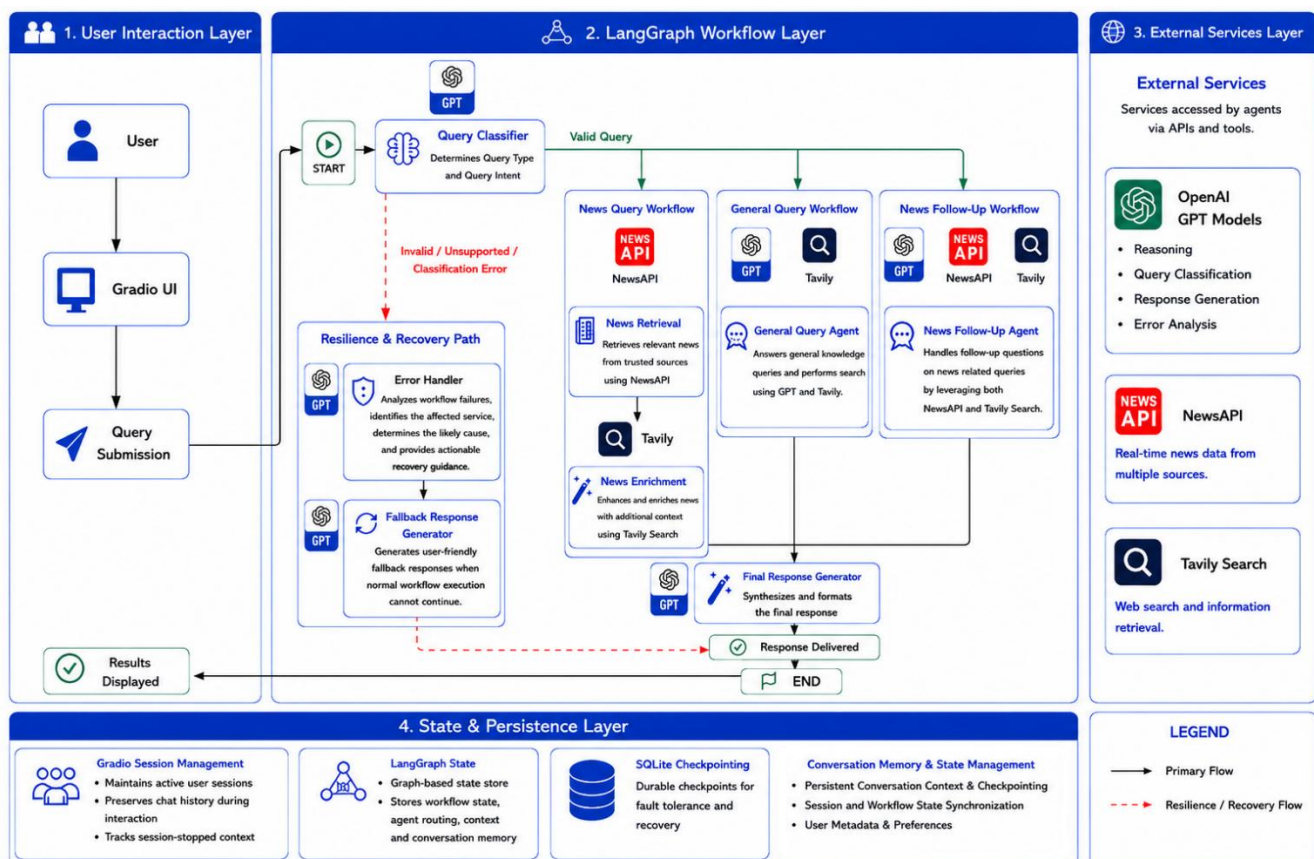


Figure 1: NewsGenie Agentic AI Workflow Architecture Diagram

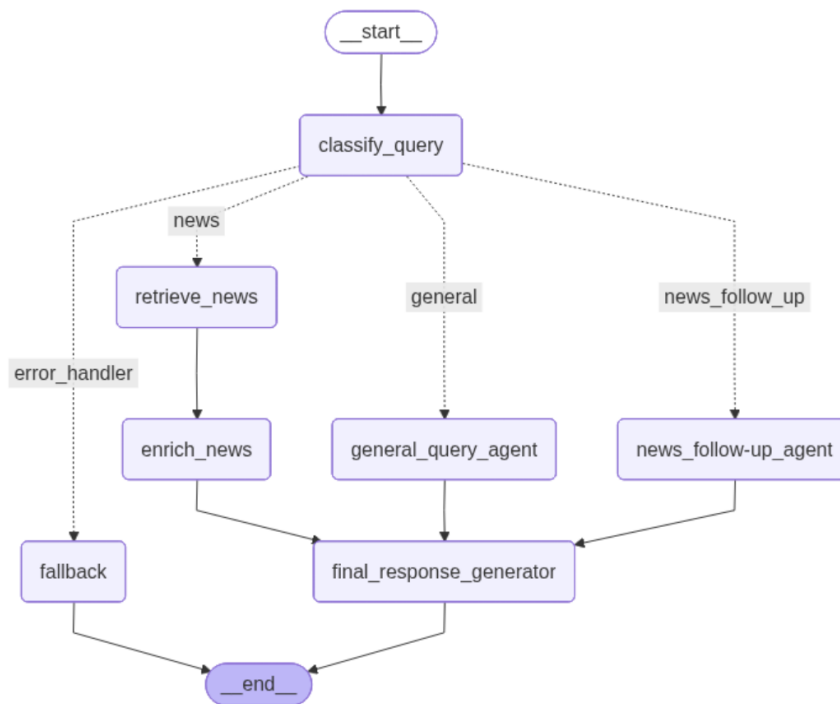


Figure 2: NewsGenie LangGraph Orchestration Workflow

Functional Component Overview

User Interaction Layer

- **Purpose**
 - Acts as the entry and exit point for all user interactions with NewsGenie.
- **Components**
 1. User – Initiates news, search, and general knowledge requests.
 2. Gradio UI – Provides an intuitive chat-based interface for query submission and response presentation.
 3. Query Submission – Captures user input and forwards requests to the LangGraph orchestration layer.
 4. Results Display – Presents generated responses back to the user in a conversational format.
- **Business Value**
 - Provides a simple and unified experience for accessing both news and knowledge services.

LangGraph Workflow Layer

- **Purpose**
 - Acts as the orchestration and decision-making engine of NewsGenie.
- **Components**
 - Query Classifier:
 - ✓ Determines Query Type & Query Intent
 - ✓ Supported Classifications – New News Query, New General Query, Follow-up News Query, Follow-up General Query, API/Key Error Detection for Fallback
 - ✓ Enablers – OpenAI GPT LLM
 - News Query Workflow:
 - ✓ Responsible for: -
 - News Retrieval – Retrieves relevant news articles from trusted sources using NewsAPI.
 - News Enrichment – Enhances retrieved news with additional context using Tavily Search.
 - ✓ Enablers – NewsAPI, Tavily Search

- General Query Workflow:
 - ✓ Handles: -
 - General knowledge questions
 - Information lookup requests
 - Search-assisted responses
 - ✓ Enablers – OpenAI GPT LLM, Tavily Search
- News Follow-Up Workflow:
 - ✓ Processes follow-up questions related to previously retrieved news articles and Contextual News Analysis requests
 - ✓ Enablers – NewsAPI, Tavily Search, OpenAI GPT LLM
- Final Response Generator:
 - ✓ Consolidates Retrieved information / Search results / Generated insights and produces a well-formatted & structured user-facing response.
 - ✓ Enablers – OpenAI GPT LLM
- Resilience & Recovery Path:
 - ✓ Error-handler (Fallback) that diagnoses API failures due to missing keys or any other API-call failures & workflow-exceptions; and, produces user-friendly responses when standard workflow execution cannot continue.
 - ✓ Enablers – OpenAI GPT LLM
- **Business Value**
 - Ensures intelligent routing, efficient workflow execution and resilient service delivery.

External Services Layer

- **Purpose**
 - Provides external intelligence and information sources used by NewsGenie.
- **Components**
 1. OpenAI GPT Large Language Model – Used for Query classification, Reasoning, Output generation, Final response generation & Error analysis
 2. NewsAPI – Provides real-time news headlines retrieval, category-based news access from trusted sources.
 3. Tavily Search – Provides Web-search capabilities, Information enrichment & Supplement context retrieval.
- **Business Value**
 - Enables access to current, accurate and contextual information beyond the internal workflow.

State & Persistence Layer

- **Purpose**
 - Maintains Persistent conversation context, Session synchronization & User preferences (also metadata).
- **Components**
 - Gradio Session Management
 - ✓ Maintains Active user sessions, Chat history & User interaction context.
 - LangGraph State
 - ✓ Stores Workflow state, Routing decisions & Conversation context.
 - SQLite Checkpointing
 - ✓ Provides Durable state persistence, Recovery support & Fault tolerance.
- **Business Value**
 - Enables context-aware conversations and reliable workflow recovery.

LangGraph Workflow Explanation

Gradio User Interface

- **Purpose**
 - The Gradio UI serves as the interaction layer between the user and the NewsGenie platform. It provides a conversational interface for submitting requests, viewing responses and maintaining interaction continuity across multiple exchanges.
- **Key Responsibilities**
 - Query & News Category Capture
 - ✓ Accepts natural language input from users.
 - ✓ Supports multiple query types including news requests, general knowledge questions and follow-up interactions.
 - ✓ Acts as the entry point into the LangGraph workflow.
 - Conversation Management
 - ✓ Maintains chat history throughout the active session.
 - ✓ Preserves prior user interactions for contextual processing.
 - ✓ Enables seamless multi-turn conversations.
 - Request Packaging
 - ✓ Combines the current user query with existing conversation history.
 - ✓ Formats inputs into the structure expected by the LangGraph workflow.
 - ✓ Ensures all required metadata is available before workflow execution begins.
 - Response Rendering
 - ✓ Displays generated responses in a conversational format.
 - ✓ Presents news summaries, enriched content and follow-up responses consistently.
 - ✓ Provides a user-friendly abstraction over the underlying workflow complexity.
 - Session Management Strategy
 - ✓ Session Persistence – Each user interaction is associated with an active session. Previous exchanges remain accessible throughout the conversation lifecycle. Session data is preserved across workflow executions.
- **Data Passed to LangGraph**
 - Current User Query & News Category
 - Conversation History & Session Context through Session ID

Memory and Persistence Framework

- **Purpose**
 - The Memory and Persistence Framework enables NewsGenie to maintain conversational continuity, preserve workflow state and support context-aware interactions across multiple user requests.
 - The framework is implemented through three complementary components: Gradio Session Management, LangGraph State Management and SQLite-based Checkpointing: -
 1. **Gradio Session Management**
 - Acts as the conversation entry point and maintains the active user session throughout the interaction lifecycle.
 - Generate and maintain a unique Session ID for each user conversation.
 - Capture user queries and conversation history.
 - Preserve chat interactions during the active session.
 - Pass session context to the LangGraph workflow.
 - When a user submits a query – Session ID is retrieved from the active Gradio session. Conversation history is collected. User query and session context are packaged and forwarded to

LangGraph. Session ID is passed as part of the workflow configuration to ensure subsequent requests are associated with the correct conversation thread.

2. LangGraph State Management

- Acts as the central working memory layer during workflow execution.
- Maintain state throughout graph execution.
- Share information between workflow nodes.
- Store intermediate and final processing results.
- Track workflow progression and routing decisions.
- The state object is initialized when a request enters the workflow and is continuously updated as execution progresses through the various nodes.
- The NewsGenie workflow uses a custom Graph State model that contains fields corresponding to the major processing stages of the workflow: -

State Variable	Description	Used By
user_query	Stores the original user request received from the Gradio interface.	All workflow nodes
news_category	Stores the selected or inferred news category (Technology, Sports, Finance, etc.).	News Retrieval Node
query_type	Stores the classifier output indicating the workflow path to be executed.	Routing Logic
query_type_hint	Stores additional classification hints used to improve routing decisions.	Query Classifier
query_intent	Stores the identified user intent used for downstream processing.	Workflow Routing
classifier_reasoning	Stores the reasoning generated by the classifier explaining the routing decision.	Auditability & Debugging
combined_news_results	Stores consolidated news articles retrieved from NewsAPI.	News Retrieval & Enrichment Nodes
enriched_news_results	Stores news articles enriched with additional context from Tavily Search.	Response Generation
general_query_response	Stores responses generated by the General Query Agent.	Response Generation
final_response	Stores the final user-facing response returned by the workflow.	Final Response Generator
error	Indicates whether workflow execution encountered an error.	Error Handler
error_source	Stores the node or component that generated the error.	Error Handler
error_message	Stores detailed error information.	Error Handler
llm_error	Indicates whether the failure originated from an LLM interaction.	Error Handler

Table 1: Graph State

- In addition to the explicit Graph State variables, the workflow leverages LangGraph's **Message State** capability. The implementation uses an annotated **hidden state variable**: **'messages'**, that stores the conversation history exchanged between the user and assistant throughout the session. Unlike the explicit Graph State fields, the 'message' state is automatically maintained by LangGraph and is not manually updated by individual workflow nodes.

3. LangGraph SQLite Checkpointing

- Provides durable persistence of workflow state and conversation context beyond a single execution cycle.
- Persist workflow checkpoints to SQLite.
- Store graph execution state.
- Maintain conversation continuity across requests.
- Enable recovery of previous workflow context.
- Checkpointing Process: -
 - ✓ During workflow execution, LangGraph periodically captures the current graph state.
 - ✓ State snapshots are written to SQLite storage.
 - ✓ Checkpoints are associated with the corresponding Session ID.
 - ✓ Previous checkpoints can be retrieved when the same user continues the conversation.
- Context Restoration: -
 - ✓ When a new query is received, Session ID is used to locate existing checkpoints.
 - ✓ Previous graph state is restored.
 - ✓ Conversation history is reconstructed.
 - ✓ Workflow execution resumes with full contextual awareness.

– State Lifecycle

- **Workflow Initialization**
 - ✓ When a user submits a request, Session ID is received from Gradio Session Management.
 - ✓ Existing conversation context is retrieved using SQLite checkpoints.
 - ✓ Graph State is initialized.
 - ✓ Previous Message State is restored.
- **During Workflow Execution**
 - ✓ Each node reads required state variables.
 - ✓ Updates relevant outputs.
 - ✓ Persists intermediate processing results.
 - ✓ Contributes to the evolving workflow context.
- **Workflow Completion**
 - ✓ Upon successful completion, final Response is written to state.
 - ✓ Updated Message State is persisted.
 - ✓ Workflow checkpoint is stored in SQLite.
 - ✓ Session context becomes available for future interactions.

Query Classification Node

– Purpose

- The Query Classification Node serves as the primary orchestration gateway for the NewsGenie workflow. Beyond determining query type and intent, the node is responsible for validating external dependencies, reconstructing conversation context, classifying requests and propagating workflow errors before downstream processing begins.
- As the first operational node in the LangGraph workflow, it acts as both a decision engine and a workflow readiness validation layer.
- The classification result directly controls workflow routing and ensures that requests are processed by the most appropriate downstream agent.

- **Inputs to Node**
 - The node receives the following information from Graph State:
 - ✓ **user_query** – Current user request
 - ✓ **messages** – Historical conversation context maintained through Message State
 - ✓ **session_id** – Session identifier passed from Gradio
 - ✓ **Previous Graph State** – Retrieved through LangGraph checkpoint restoration
- **Step 1 – API Dependency Validation**
 - Before performing any query classification, the node executes a series of pre-validation checks to ensure that all required external dependencies are available and operational. This proactive validation approach prevents the workflow from progressing when downstream services are unavailable, thereby avoiding unnecessary processing and expensive LLM invocations.
 - **API Key Validation – OpenAI GPT, NewsAPI & Tavily**
 - ✓ Verifies the presence and accessibility of all mandatory API credentials required by the solution.
 - ✓ Detects missing, invalid or incorrectly configured API keys before workflow execution begins.
 - **API Connectivity & Service Validation**
 - ✓ Validates external services (OpenAI GPT, NewsAPI & Tavily) are reachable and responding correctly.
 - ✓ Performs lightweight authentication and connectivity checks against dependent services.
 - ✓ Detects failures caused by:
 - Invalid API credentials
 - Expired or revoked access tokens
 - Service unavailability
 - Network or connectivity issues
 - Configuration errors
 - **Error Propagation Strategy**
 - ✓ Any validation failure immediately updates the workflow state with: Error Flag, Error Source, Error Message
 - ✓ Prevents execution of downstream workflow nodes.
 - ✓ Routes the request directly to the Error Handler and Fallback workflow, ensuring a graceful user experience.
- **Step 2 – Classification Strategy**
 - The implementation uses a **hybrid classification approach** combining deterministic rules with LLM-based reasoning.
 1. **Stage 1 – Deterministic Classification**
 - ✓ Before invoking the LLM, the workflow executes a helper function that analyses the user query using predefined keyword patterns.
 - ✓ The objective is to generate a preliminary classification hint: news / general / none
 - ✓ This hint is used to improve classification consistency and reduce unnecessary ambiguity during LLM evaluation.
 2. **Stage 2 – LLM-Based Classification**
 - ✓ The query is then evaluated using an LLM-as-Judge Classifier.
 - ✓ Unlike traditional intent detection, the classifier evaluates two independent dimensions simultaneously: Query Intent (new_topic / follow_up) & Query Type (news / general).
 - ✓ This enables the workflow to distinguish not only what the user wants but also whether the request depends on previous conversation context.

- **Prompt Engineering Strategy**
 - ✓ The classifier prompt defines a dedicated system role.
 - ✓ Inputs to the Classifier – Current User Query, Conversation History, Query Type Hint
 - ✓ Determine Query Intent first (new_query / follow_up)
 - ✓ Determine Query Type second (news / general)
 - ✓ Consider conversation history
 - ✓ Consider deterministic type hints
 - ✓ Return structured outputs only
 - ✓ Avoid answering the user's question
 - ✓ Strongly favour the deterministic type hint when provided.
 - ✓ Override the hint only when the query clearly indicates otherwise.
 - ✓ Never use the hint while determining conversational intent.
- **Structured Output Enforcement**
 - ✓ To eliminate free-form classifier responses, the implementation uses a Pydantic schema and LangChain structured output binding.
 - ✓ The classifier is constrained to return:
 1. **query_type** – news or general
 2. **query_intent** – new_topic or follow_up
 3. **classifier_reasoning** – Explanation of classification decision

Conditional Routing Logic

- **Purpose**
 - The Conditional Routing Layer evaluates the classification results generated by the Query Classification Node and dynamically directs the request to the appropriate workflow branch.
- **Routing Inputs**
 - The routing decision is based on the following state variables:
 - ✓ **query_type** – Identifies whether the request is a News or General query
 - ✓ **query_intent** – Identifies whether the request is a New Topic or Follow-Up interaction
 - ✓ **error** – Indicates whether a validation or processing error has occurred
- **Routing Logic**
 - The workflow follows a simple hierarchical routing model: -

Condition	Next Node
error = True	Error Handler
query_type = general	General Query Agent
query_type = news and query_intent = new_topic	News Retrieval Workflow
query_type = news and query_intent = follow_up	News Follow-Up Agent
Default	General Query Agent

Table 2: Routing Logic

- **Design Considerations**
 - Error conditions are evaluated first to prevent unnecessary workflow execution.
 - New news requests are routed through the retrieval and enrichment pipeline.
 - Follow-up news requests leverage previously retrieved news and conversation context.
 - General knowledge requests bypass news retrieval services and are handled directly by the General Query Agent.
 - A default route is maintained to ensure all requests receive a response even if classification results are incomplete.

News Query Workflow

- **Purpose**
 - The News Query Workflow is responsible for retrieving, enriching and consolidating real-time news content before generating the final response. This workflow is triggered when the Query Classification Node identifies the request as a **News Query** with an intent of **New Topic**.
- **Step 1 – News Retrieval**
 - Retrieve comprehensive and relevant news coverage by combining both category-based and query-specific retrieval strategies.
 - To improve both coverage and relevance, the workflow performs two independent NewsAPI retrieval operations: -
 - ✓ **Category-Based Retrieval** – Retrieves the latest headlines for the user-selected news category (Technology, Business, Sports, Finance, etc.).
 - ✓ **Query-Based Retrieval** – Retrieves articles specifically related to the user's query to improve topical relevance.
 - The outputs from both retrieval operations are **merged** into a unified news dataset. During consolidation, the workflow: -
 - ✓ Combines results from both NewsAPI calls.
 - ✓ Removes duplicate articles and overlapping coverage.
 - ✓ Creates a consolidated news package for downstream processing.
 - Status Updates: -
 - ✓ **combined_news_results** – Stores the merged & de-duplicated results from category-based and query-based news retrieval.
 - Design Benefits: -
 - ✓ This dual-retrieval strategy ensures the solution captures major developments within the selected news category & user-specific news topics and events
 - ✓ Broader coverage while maintaining query relevance.
 - ✓ Reduced risk of missing important stories due to reliance on a single retrieval method.
- **Step 2 – News Enrichment**
 - Enhance retrieved news articles with additional context and supporting information.
 - The workflow invokes Tavily Search to enrich the retrieved news articles with complementary information from external sources.
 - Key activities include: -
 - ✓ Using article topics as enrichment queries.
 - ✓ Retrieving supporting information additional context / background information from trusted web sources
 - ✓ Supplementing NewsAPI results where article summaries are limited.
 - Status Updates: -
 - ✓ **enriched_news_results** – Stores news articles enriched with external context and supporting information.
 - Design Benefits: -
 - ✓ Improves information depth beyond headline-level reporting.
 - ✓ Provides additional context for complex or developing stories.
 - ✓ Reduces information fragmentation by presenting a unified view of the topic.
- **Error Handling**
 - The General Query Workflow includes exception handling to capture runtime failures. When an exception occurs, the node updates the workflow state with the error flag, source and detailed error message, enabling consistent error tracking and diagnostics across the workflow.

General Query Workflow

- **Purpose**
 - The General Query Workflow is responsible for handling informational and knowledge-based requests that do not require real-time news retrieval. This workflow is triggered when the Query Classification Node identifies the request as a **General Query**.
- **General Query ReAct Agent**
 - Provide accurate, factual and trustworthy responses to user queries while leveraging external search capabilities when additional information is required.
 - The workflow utilizes a dedicated **ReAct Agent** configured with **Tavily** Search access.
 - Unlike the News Workflow, this branch does not invoke NewsAPI and focuses solely on answering the user's question directly.
 - Key activities include: -
 - ✓ Interpreting the user's question.
 - ✓ Determining whether external information is required.
 - ✓ Invoking Tavily Search when additional context or current information is needed.
 - ✓ Combining retrieved information with LLM reasoning.
 - ✓ Generating a structured response for downstream processing.
- **Prompt Engineering Strategy**
 - The General Query Agent is governed by a dedicated system prompt that instructs the model to: -
 - ✓ Act as NewsGenie's General Assistant.
 - ✓ Provide factual and trustworthy responses.
 - ✓ Utilize external search tools when required.
 - ✓ Prioritize accuracy over speculation.
 - ✓ Generate clear and concise explanations.
- **Status Updates**
 - **general_query_response** – Stores the response generated by the ReAct Agent along with supporting context.
- **Design Benefits**
 - Supports a broad range of informational and knowledge-based queries.
 - Enables access to current external information through Tavily Search.
 - Reduces hallucinations by grounding responses in retrieved information when required.
 - Provides a flexible framework capable of answering both static and dynamic user questions.
- **Error Handling**
 - The General Query Agent includes exception handling to capture runtime failures. When an exception occurs, the node updates the workflow state with the error flag, source and detailed error message, enabling consistent error tracking and diagnostics across the workflow.

News Follow-Up Workflow

- **Purpose**
 - The News Follow-Up Workflow handles contextual questions related to previously discussed news topics. This workflow is triggered when the Query Classification Node identifies the request as a **News Query** with an intent of **Follow-Up**.
 - Unlike the News Query Workflow, which follows a predefined retrieval and enrichment pipeline, this workflow leverages a dedicated **ReAct Agent** capable of reasoning over conversation context and dynamically selecting the appropriate tools required to answer the user's question.
- **News Follow-Up ReAct Agent**
 - Provides contextual, research-oriented responses to questions related to previously discussed news topics.
 - The workflow utilizes a dedicated **ReAct Agent** configured with access to both **NewsAPI** and **Tavily Search** tools.
 - Unlike the News Query Workflow, which follows a fixed retrieval path, this agent dynamically determines the information required to answer the user's follow-up question.
 - The agent leverages conversation history maintained through LangGraph memory to understand contextual references and user intent.
 - Key Activities: -
 - ✓ Interpreting the user's follow-up question in the context of prior interactions.
 - ✓ Resolving ambiguous references
 - ✓ Determining whether additional information is required from NewsAPI & Tavily Search
 - ✓ Retrieving factual updates and supporting contextual information.
 - ✓ Combining retrieved information with LLM reasoning.
 - ✓ Generating a focused, research-backed response for downstream processing.
- **Prompt Engineering Strategy**
 - The News Follow-Up Agent is governed by a dedicated system prompt designed specifically for contextual news conversations.
 - The prompt instructs the model to: -
 - ✓ Analyse the user's question in conjunction with conversation history.
 - ✓ Reconstruct incomplete or ambiguous follow-up requests into meaningful research queries.
 - ✓ Utilize NewsAPI and Tavily Search when additional information is required.
 - ✓ Gather both factual updates and contextual insights before generating a response.
 - ✓ Answer the user's question directly rather than generating generic news summaries.
 - ✓ Maintain a factual, analytical and conversational tone.
 - The prompt also guides the agent to differentiate between:
 - ✓ News Updates – Recent developments and factual events.
 - ✓ Contextual Analysis – Background information, implications, comparisons, and industry impact.
 - This approach enables the workflow to move beyond simple article retrieval and provide research-oriented responses tailored to the user's specific question.
- **Design Benefits**
 - Improves interpretation of contextual follow-up questions.
 - Provides flexibility through dynamic tool selection.
 - Enhances retrieval accuracy through query reformulation.
 - Produces more relevant and targeted responses.
 - Supports conversational exploration of news topics.

- **State Updates**
 - **general_query_response** – Stores the response generated by the News Follow-Up ReAct Agent for downstream response generation.
- **Error Handling**
 - The General Query Agent includes exception handling to capture runtime failures. When an exception occurs, the node updates the workflow state with the error flag, source and detailed error message, enabling consistent error tracking and diagnostics across the workflow.

Final Response Generator

- **Purpose**
 - The Final Response Generator serves as the final orchestration layer within the NewsGenie workflow. Its responsibility is to consolidate outputs generated by upstream workflow branches and transform them into a clear, user-friendly response.
 - This node ensures that responses are presented consistently regardless of whether the request originated from the News Query Workflow, General Query Workflow or News Follow-Up Workflow.
- **Context Selection Logic**
 - Before generating the final response, the node selects the appropriate context source based on the workflow path executed: -

Workflow Path	Context Used
News Query + New Topic	enriched_news_results
General Query	general_query_response
News Follow-Up Query	general_query_response

Table 3: Context Selection Logic | Final Response Generator Node

- This ensures that the response generator consumes the most relevant information produced by upstream workflow nodes before generating the final user-facing response.
- **Final Response Generation**
 - Utilizes a dedicated LLM chain with structured output enforcement (string).
 - Consumes: User Query, Query Type, Query Intent, News Category, Workflow Context
 - Transforms retrieved and enriched information into a coherent response.
 - Removes duplicate information and unnecessary verbosity.
 - Produces a single user-facing response irrespective of the workflow path executed.
 - Key Activities: -
 - ✓ Consolidating outputs from upstream workflow nodes.
 - ✓ Normalizing response structure across workflow branches.
 - ✓ Removing duplicate content.
 - ✓ Converting research results into conversational responses.
 - ✓ Generating the final response returned to the user.
- **Prompt Engineering Strategy**
 - The Final Response Generator is governed by a dedicated system prompt that positions the model as NewsGenie's News and Research Assistant.
 - Based on these inputs, the model **dynamically assumes different response-generation roles**:
 - ✓ **News + New Topic** – News Analyst responsible for generating structured and informative news briefings from enriched news content.
 - ✓ **News + Follow-Up** – Research Assistant responsible for answering contextual questions using retrieved news and supporting research.

- ✓ **General Query** – Knowledge Assistant responsible for providing factual and trustworthy responses to informational requests.
- The prompt instructs the model to:
 - ✓ Generate responses using only the supplied context.
 - ✓ Generate responses only from information originating from articles with valid source attribution, helping reduce misinformation and unsupported claims.
 - ✓ Treat the provided context as the single source of truth.
 - ✓ Never hallucinate, speculate, infer, or fabricate information.
 - ✓ Exclude information not present within the supplied context.
 - ✓ Remove duplicate information
 - ✓ Remove tool traces, agent reasoning, intermediate steps, and workflow details.
 - ✓ Present information in a clear, conversational and user-friendly manner based on the assumed role (News Format vs Conversational Format)
 - ✓ This approach enables a single response generation layer to support multiple interaction patterns while maintaining consistency, factual accuracy, and a natural conversational experience.
- **Design Benefits**
 - Provides workflow-specific response generation behaviour.
 - Ensures news queries and research queries are presented differently.
 - Maintains consistency across all workflow branches.
 - Prevents exposure of internal agent reasoning and execution details.
 - Improves response trustworthiness by ensuring generated responses are grounded in source-attributed news content.
 - Improves overall response quality and user experience.
- **Error Handling**
 - The General Query Agent includes exception handling to capture runtime failures. When an exception occurs, the node updates the workflow state with the error flag, source and detailed error message, enabling consistent error tracking and diagnostics across the workflow.

Error Handling and Recovery Workflow

- **Purpose**
 - The Error Handling and Recovery Workflow provides a centralized mechanism for managing failures across the NewsGenie application. Rather than allowing individual workflow nodes to fail silently or expose technical exceptions to users, all errors are standardized and routed through a dedicated recovery workflow.
- **Error State Management**
 - All workflow nodes implement exception handling and update a common error structure within Graph State when a failure occurs: -
 - ✓ **error** – Indicates whether a workflow failure has occurred
 - ✓ **error_source** – Identifies the component where the error originated
 - ✓ **error_message** – Captures the detailed exception message
 - ✓ **llm_error** – Indicates whether the failure originated from an LLM invocation
 - This standardized structure allows errors from any workflow branch to be processed consistently.
- **LLM Failure Handling**
 - A special handling mechanism exists for LLM-related failures. When an LLM invocation fails, the originating node updates: **llm_error = True**
 - Since the Error Handler itself relies on an LLM, it cannot be invoked during an LLM outage or authentication failure.

- In these scenarios, the workflow bypasses the Error Handler and returns a predefined fallback response indicating that the language model service is currently unavailable and advising the user to retry later.
- This prevents cascading failures and ensures that users always receive a response, even when the primary reasoning engine is unavailable.

– Error Handler

- The workflow utilizes a dedicated Error Handler to process failures captured within Graph State.
- The Error Handler receives: User Query, Error Source, Error Message
- Rather than exposing raw technical exceptions, it generates user-friendly error explanations and recovery guidance.
- Key Activities: -
 - ✓ Explaining what failed.
 - ✓ Translating technical errors into plain language.
 - ✓ Identifying the likely cause of the issue.
 - ✓ Recommending actionable next steps.
 - ✓ Guiding users towards alternative actions where appropriate.
- Prompt Engineering Strategy
 - ✓ The Error Handler is governed by a dedicated system prompt that positions the model as NewsGenie's Error Analysis Agent.
 - ✓ The prompt instructs the model to: -
 - Explain what failed in clear, non-technical language.
 - Describe the most likely reason for the failure.
 - Provide practical recovery recommendations.
 - Suggest alternative actions when available.
 - Maintain a helpful and reassuring tone.
 - Focus on enabling the user to successfully complete their task.
 - ✓ Every response is designed to answer three key questions:
 - What failed?
 - Why did it fail?
 - What should the user do next?
 - ✓ This approach transforms technical failures into guided recovery experiences rather than generic error messages.

– Design Benefits

- Provides a centralized mechanism for handling errors across all workflow branches.
- Ensures consistent error tracking through standardized Graph State variables.
- Converts technical exceptions into user-friendly recovery guidance.
- Simplifies troubleshooting by capturing error source and detailed error messages.
- Improves application resilience by preventing workflow failures from propagating downstream.
- Maintains a consistent user experience during API, tool, or processing failures.
- Includes dedicated handling for LLM outages to prevent cascading failures when the primary reasoning engine is unavailable.

Fallback Strategy

- The solution incorporates multiple fallback mechanisms to ensure users continue to receive meaningful responses even when external dependencies fail or information retrieval is incomplete.
- **Technical Fallbacks**
 - **API Key & Authentication Failures**
 - ✓ User Interface doesn't allow empty API keys. Chat functionality hidden if any API key is missing.
 - ✓ Missing, invalid, or expired API credentials are detected during the Query Classification stage.
 - ✓ The workflow is halted before any downstream processing is initiated.
 - ✓ Error details are captured and passed to the Error Handler for user-friendly guidance.
 - **API Connectivity & Runtime Failures**
 - ✓ Failures during NewsAPI, Tavily or workflow execution are captured using standardized Graph State variables.
 - ✓ The workflow routes the request to the Error Handler, which explains the issue and recommends recovery actions.
 - **LLM Failure**
 - ✓ LLM invocation failures are identified using the llm_error flag.
 - ✓ Since all intelligent agents rely on the LLM, the Error Handler cannot be invoked during an LLM outage.
 - ✓ In such scenarios, the workflow returns a predefined fallback message informing the user that the AI service is temporarily unavailable.
- **Workflow-Level Fallbacks**
 - **News Query Workflow**
 - ✓ News content is retrieved using both category-based and query-based NewsAPI searches.
 - ✓ If one retrieval strategy returns limited results, the second retrieval path improves coverage.
 - ✓ Retrieved articles are further enriched using Tavily Search to supplement missing context and supporting information.
 - **News Follow-Up Workflow**
 - ✓ The ReAct Agent has access to both NewsAPI and Tavily Search.
 - ✓ If sufficient information is not available from news sources, the agent can leverage Tavily Search to gather additional context, analysis, and supporting information.
 - **General Query Workflow**
 - ✓ The ReAct Agent utilizes Tavily Search when additional information is required.
 - ✓ This allows the workflow to supplement LLM knowledge with external information and provide more complete responses.
- **Design Benefits**
 - Reduces dependency on a single API or retrieval strategy.
 - Improves resilience against service outages and runtime failures.
 - Maximizes information coverage through multiple retrieval paths.
 - Maintains a consistent user experience during degraded operating conditions.
 - Ensures users receive the most relevant response possible even when primary information sources are unavailable.

Section 3 – Gradio User Interface

User Interface Screenshot

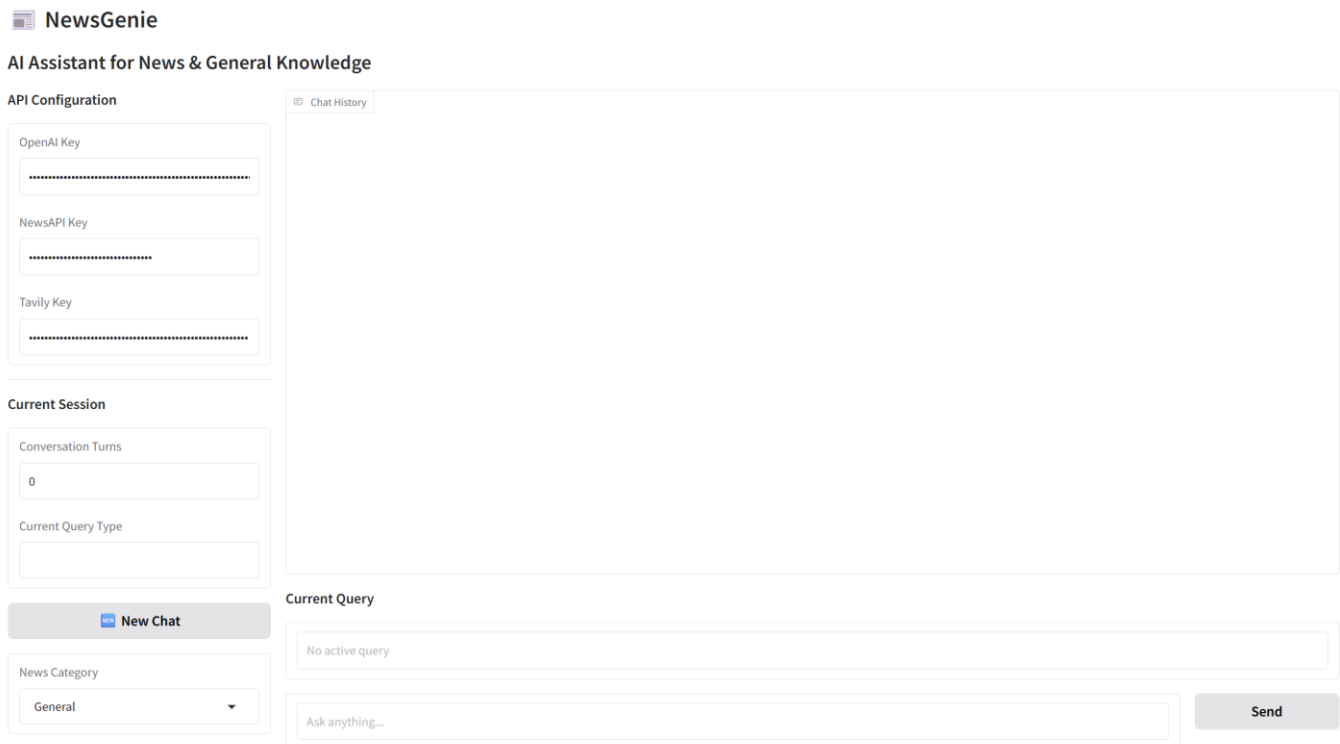


Figure 3: NewsGenie User Interface Screenshot | Gradio UI

User Interface Overview

- Single-screen interface for all interactions.
 - Dynamic API configuration without code changes.
 - Persistent conversation history within the active session.
 - Support for multi-turn and follow-up conversations.
 - Real-time query submission and response generation.
 - Session reset capability through the New Chat function.
 - This lightweight interface provides a user-friendly front end for interacting with the underlying LangGraph-powered multi-agent workflow.
- The UI is divided into four primary sections: -
 1. API Configuration Panel
 - Located on the left-hand side, this section allows users to provide the API credentials required by the application.
 - ✓ **OpenAI Key** – Used for query classification, agent reasoning, response generation, and error analysis.
 - ✓ **NewsAPI Key** – Used to retrieve real-time news articles and headlines.
 - ✓ **Tavily Key** – Used for web search, contextual enrichment, and research-oriented responses.

- This section enables dynamic configuration of external services without modifying application code.

2. Session Information Panel

- This section displays information related to the current conversation session.
- ✓ **Conversation Turns** – Displays the number of interactions that have occurred within the active session.
- ✓ **Current Query Type** – Displays the most recent query classification result in the format <Query_Type> (<Query_Intent>). For eg., news (follow_up)
- ✓ **New Chat** – Button that resets the current conversation and starts a new session.
- This information helps users understand the current conversation state and maintain awareness of the active workflow context.

3. Chat History Panel

- The central area of the interface is dedicated to displaying the conversation history.
- Features include:
 - ✓ Displays all user and assistant interactions for the current session.
 - ✓ Maintains conversational context across multiple turns.
 - ✓ Enables users to review previously retrieved news articles and responses.
 - ✓ Supports follow-up interactions by preserving conversation history.

4. Query Submission Panel

- Located at the bottom of the interface, this section allows users to submit new requests.
 - ✓ **Current Query** – Displays the currently active query being processed.
 - ✓ **News Category** – Dropdown menu used to select a preferred news category (e.g., Technology, Business, Sports, Finance, General).
 - ✓ **Query Input Box** – Text input area where users enter questions or requests.
 - ✓ **Send Button** – Submits the query to the LangGraph workflow for processing.

API Missing Key UI Validation

- The chat section disappears if API key is missing and the following message is displayed: -

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

Configure API Keys

Please provide:

- OpenAI API Key
- NewsAPI Key
- Tavily API Key

The chat interface will appear automatically.

Current Session

Conversation Turns: 0

Current Query Type:

New Chat

News Category: General

Figure 4: API Missing Key UI Validation

Section 4 – Sample Output Screenshots

Technology News

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

10

Current Query Type

news (new_topic)

New Chat

News Category

Technology

Chat History

Latest Technology headlines

Executive Summary

The latest technology headlines highlight significant developments, including a large-scale malware attack on Arch Linux's User Repository, the innovative use of generative AI in the development of "Tomb Raider: Legacy of Atlantis," and the upcoming launch of Valve's Steam Frame headset this summer.

Key Developments

- 1. Arch Linux's AUR Compromised**
 - What Happened:** Over 400 packages in the Arch Linux User Repository were compromised in a malware campaign. Arch Linux maintainers are actively working to remove the malicious content and ban affected accounts.
 - Why It Matters:** This incident raises concerns about the security of user-supplied packages in open-source repositories and highlights the need for vigilant package management.
 - Source:** Phoronix, [Read more](#) (Published on June 12, 2026).
- 2. Use of Generative AI in "Tomb Raider: Legacy of Atlantis"**
 - What Happened:** Developers Crystal Dynamics and Flying Wild Hog utilized generative AI to create placeholder assets during the game's development, speeding up the ideation process while ensuring that all final content remains human-crafted.
 - Why It Matters:** This approach showcases how AI can enhance creativity in game development without compromising artistic integrity, a topic of growing interest in the gaming community.

Current Query

Latest Technology headlines

Ask anything...

Send

Figure 5: Technology News Screenshot

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

12

Current Query Type

news (follow_up)

New Chat

News Category

Technology

Chat History

Elaborate on GenAI usage in Tomb Raider

GenAI, or Generative AI, has been increasingly integrated into the gaming industry, including in popular franchises like Tomb Raider. This technology is being utilized in various aspects of game development and player experience, enhancing storytelling, character development, and environmental design.

Key Uses of GenAI in Tomb Raider:

- 1. Dynamic Storytelling:** GenAI can help create more adaptive narratives that respond to player choices, making the gameplay experience more immersive and personalized. This allows for branching storylines that can evolve based on player interactions.
- 2. Character Development:** AI-generated dialogue and character interactions can lead to richer character arcs and more believable interactions. This enhances the emotional depth of the game, making players feel more connected to the characters.
- 3. Procedural Content Generation:** GenAI can assist in designing expansive game worlds by generating diverse landscapes, quests, and challenges. This not only saves time for developers but also ensures that players encounter unique experiences each time they play.
- 4. Enhanced Graphics and Animation:** AI can be used to improve graphics rendering and character animations, making the visual experience more lifelike and appealing.
- 5. Player Engagement:** By analyzing player behavior and preferences, GenAI can tailor experiences to keep players engaged, offering them challenges and storylines that resonate with their interests.

Implications for the Franchise:

Current Query

Elaborate on GenAI usage in Tomb Raider

Ask anything...

Send

Figure 6: News Follow-up Query (on Tech. News) Screenshot

Finance News

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

13

Current Query Type

news (new_topic)

New Chat

News Category

Finance

Chat History

Finance News

Executive Summary

Recent finance news highlights significant developments in the market, particularly surrounding the highly anticipated SpaceX IPO, which is set to become the largest in history. This event has positively influenced stock indexes, with the Dow, S&P 500, and Nasdaq showing gains as investors react to the market dynamics.

Key Developments

- SpaceX IPO Launch:** SpaceX is debuting on the stock market with shares priced at \$135, expected to open at around \$150 or higher. The IPO is anticipated to raise substantial capital, marking a historic moment for the company and the market. **Why It Matters:** This IPO is significant not just for SpaceX but for the broader market, as it reflects investor confidence and could lead to increased activity in related sectors. **Source:** The Wall Street Journal, [WSJ Article](#), June 12, 2026.
- Market Reactions:** Following the IPO news, the Dow Jones Industrial Average rose by 353 points, while the S&P 500 and Nasdaq also saw increases. This rally is attributed to investor optimism around SpaceX's performance and potential future growth. **Why It Matters:** The rise in these indexes indicates a strong market sentiment, which can lead to further investment and economic growth. **Source:** CNBC, [CNBC Article](#), June 12, 2026.

Current Query

Finance News

Ask anything...

Send

Figure 7: Finance News Screenshot

Sports News

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

14

Current Query Type

news (new_topic)

New Chat

News Category

Sports

Chat History

Sports News

Executive Summary

The U.S. Men's National Team (USMNT) is set to open its 2026 FIFA World Cup campaign against Paraguay on June 12, 2026, in Los Angeles. This match marks a significant moment as it is the first World Cup hosted on U.S. soil in over three decades, and the team features a talented roster that many consider part of the country's "golden generation" of soccer players.

Key Developments

- Match Kickoff:** The USMNT will play Paraguay in their opening match of the tournament, showcasing a roster that includes stars like Christian Pulisic and Weston McKennie. **Why It Matters:** This game is a culmination of years of preparation and development for a generation of American players who are now reaching their peak, making it a pivotal moment for U.S. soccer. **Source:** NPR, [Your cheat sheet to the 26 players on the U.S. World Cup team](#), June 12, 2026.
- Expectations and Pressure:** The team is facing high expectations as hosts, with coach Mauricio Pochettino emphasizing the belief that they can compete for the championship. **Why It Matters:** The home advantage and the presence of a supportive fanbase could significantly impact the team's performance and morale, especially as they aim to advance deep into the tournament. **Source:** NPR, [The U.S. men's team is set to take on Paraguay in its World Cup opener](#), June 12, 2026.

Current Query

Sports News

Ask anything...

Send

Figure 8: Sports News Screenshot

General & Search Queries

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

5

Current Query Type

general (new_topic)

New Chat

News Category

General

Chat History

Explain the difference between LangGraph and CrewAI in a tabular format

Here's a comparison of LangGraph and CrewAI in tabular format:

Feature/Aspect	LangGraph	CrewAI
Architecture	Graph-based with nodes and edges	Multi-agent orchestration framework
Workflow Control	Fully customizable paths, loops, and conditions	Role-based architecture for agent collaboration
State Management	Built-in persistent state across the entire graph	Agents maintain context and can operate autonomously
Support for Loops	Yes, supports cyclical flows and iterations	Not specifically designed for loops
Human-in-the-Loop	Built-in support for pausing and resuming	Supports human-in-the-loop workflows
Debugging Tools	High observability with tools like LangSmith	Advanced debugging and real-time observability
Integration	Integrates with various tools and APIs	Connects to multiple LLM providers and tools
Use Cases	Complex LLM applications and workflows	Collaborative AI agents for complex tasks

Current Query

Explain the difference between LangGraph and CrewAI in a tabular format

Ask anything...

Send

Figure 9: General Query Screenshot

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

1

Current Query Type

general (new_topic)

New Chat

News Category

General

Chat History

My name is Trump. I am from New York, US. Please share with me the current time and temperature in my city.

As of now, here is the current time and temperature in New York, US:

- Current Time: Approximately 8:47 AM EDT (Eastern Daylight Time) on June 13, 2026.
- Current Temperature: Approximately 63°F (17°C), with clear and mild conditions.

If you need more specific information or updates, feel free to ask!

Current Query

My name is Trump. I am from New York, US. Please share with me the current time and temperature in my city.

Ask anything...

Send

Figure 10: Search Query Screenshot

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

2

Current Query Type

general (follow_up)

New Chat

News Category

General

Chat History

My name is Trump. I am from New York, US. Please share with me the current time and temperature in my city.

As of now, here is the current time and temperature in New York, US:

- Current Time: Approximately 8:47 AM EDT (Eastern Daylight Time) on June 13, 2026.
- Current Temperature: Approximately 63°F (17°C), with clear and mild conditions.

If you need more specific information or updates, feel free to ask!

What is my name & city?

Your name is Trump, and you are from New York, US.

Current Query

What is my name & city?

Ask anything...

Send

Figure 11: Follow-up Query (on General Query) Screenshot

Fallback – Invalid OpenAI Key

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

1

Current Query Type

unknown (unknown)

New Chat

News Category

General

Chat History

latest news

Unable to process your request because a valid OpenAI API key is not available.

Possible causes include:

- Missing OpenAI API key
- Invalid or expired API key
- Authentication failure
- API quota or billing limitations
- OpenAI service outage or connectivity issue

Please verify your OpenAI API key configuration and try again.

If you recently updated the API key, ensure the application has been restarted or the environment variables have been refreshed.

Current Query

latest news

Ask anything...

Send

Figure 12: Fallback - Invalid OpenAI Key Screenshot

Fallback – Invalid NewsAPI Key

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

1

Current Query Type

unknown (unknown)

New Chat

News Category

General

Chat History

latest news

It looks like there was an issue retrieving the latest news because the request to the news service was unauthorized. This usually means that there's a problem with the API key you're using to access the news data.

latest news

ask anything...

Send

Current Query

latest news

Figure 13: Fallback - Invalid NewsAPI Key Screenshot

Fallback – Invalid Tavily Key

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

.....

NewsAPI Key

.....

Tavily Key

.....

Current Session

Conversation Turns

1

Current Query Type

unknown (unknown)

New Chat

News Category

General

Chat History

latest temperature in New Delhi, India

It looks like there was an issue retrieving the latest temperature in New Delhi because the request to the Tavily Search API failed due to an invalid API key. This means the system couldn't access the information it needed to provide you with the temperature data.

latest temperature in New Delhi, India

ask anything...

Send

Current Query

latest temperature in New Delhi, India

Figure 14: Fallback - Invalid Tavily Key Screenshot

Fallback – Unclear Query

NewsGenie

AI Assistant for News & General Knowledge

API Configuration

OpenAI Key

NewsAPI Key

Tavily Key

Current Session

Conversation Turns

Current Query Type

 New Chat

News Category

Chat History

give

It seems like your message was a bit unclear. Could you please provide more details about what you're looking for? Are you interested in a specific topic within entertainment, like movies, music, or celebrity news? Let me know so I can assist you better!

Current Query

give

Ask anything...

Send

Figure 15: Fallback – Unclear Query Screenshot

Section 5- Challenges Faced & Future Enhancements

Challenges Faced

Query Classification Accuracy

- One of the primary challenges was accurately distinguishing between news queries, general knowledge queries and contextual follow-up questions.
- User queries are often ambiguous and may not explicitly indicate whether the user is seeking current news or general information.
- Additional prompt engineering and contextual analysis were required to improve classification accuracy and ensure requests were routed to the correct workflow.

Follow-Up Query Interpretation

- Follow-up questions frequently contain incomplete references such as: -
 - "Tell me more about that."
 - "What happened next?"
 - "How does this compare?"
- Successfully interpreting these requests required leveraging conversation history and maintaining context across multiple interactions.
- The challenge was ensuring the system could reconstruct the user's intent without requiring them to repeatedly provide context.

Multi-Source Information Consolidation

- News information was retrieved using multiple retrieval strategies, including category-based news retrieval, query-based news retrieval and Tavily web search enrichment.
- Consolidating information from different sources while removing duplicates and maintaining response quality required additional processing and response orchestration.

Dependency on External Services

- The solution relies on multiple external services including OpenAI, NewsAPI and Tavily.
- API outages, authentication failures, rate limits, and connectivity issues can impact workflow execution.
- Robust validation, error handling and fallback mechanisms were required to ensure a consistent user experience.

Maintaining Context Across Sessions

- Preserving conversation history and workflow state across multiple interactions required careful integration of LangGraph State Management and SQLite checkpointing.
- The challenge was ensuring that follow-up questions could access relevant conversational context while maintaining workflow consistency.

Preventing Hallucinations

- Since the application combines LLM reasoning with external information retrieval, it was important to ensure responses remained grounded in retrieved information.
- Dedicated prompt engineering and response generation controls were implemented to minimize unsupported or speculative responses.

Future Enhancements

Multi-Source News Aggregation

- Integrate additional news providers alongside NewsAPI to improve coverage, reliability and source diversity.
- Enable cross-source validation of news content to improve trustworthiness.

Personalized News Recommendations

- Introduce user preferences and personalization capabilities.
- Recommend news topics based on user interests, browsing history and previous interactions.

Long-Term Memory using Vector Databases

- Extend conversational memory beyond the current session by integrating a vector database.
- Enable retrieval of historical conversations and user preferences across sessions.

Sentiment and Trend Analysis

- Incorporate sentiment analysis to identify positive, negative and neutral news trends.
- Provide insights into emerging topics and market sentiment.

Advanced News Analytics

- Provide article comparison capabilities across publishers.
- Highlight similarities, differences and conflicting viewpoints on the same news event.
- Generate trend summaries across multiple related articles.